

Isaiah Ruz

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EDUCATION

2024 - 2026	M.S. Physics at California State University, Fresno	(GPA: 3.77/4.0)
2020 - 2022	B.A. Philosophy at University of California, Santa Cruz	(GPA: 3.47/4.0)
2013 - 2017	A.S. Physics at Cuesta College	(GPA: 3.28/4.0)

RESEARCH INTERESTS

I have a strong interest in applying and developing mathematical tools like cohomology and conformal theory as well as other topological and geometric methods to theoretical physics and mathematical physics problems. The mathematical and physical aspects of conformal field theory and holographic entanglement are especially enticing to me.

RESEARCH EXPERIENCE

California State University, Fresno (January 2024-May 2024)

Project: Calculating Generalized Uncertainty Principle Wavefunction Spread

Supervisor: Douglas Singleton and Michael Bishop

Explored a possible experimental setup to verify Kempf, Mann, and Mangano's Generalized Uncertainty Approach (GUP) to Quantum Gravity. This involved modifying the position operator in quantum mechanics to obtain a new uncertainty relation with a minimal length scale. Specifically, we calculated the spread of the width of the wavefunction of molecules in ordinary quantum mechanics and compared it to the calculated width in the GUP regime.

California State University, Fresno (May 2024-May 2025)

Project: Electromagnetic Field Momentum in Theoretical Magnetic Monopole-like Models

Supervisor: Douglas Singleton

We verified the Center-of-Energy Theorem by calculating the hidden momentum and field momentum of stationary theoretical magnetic monopole-like models (Dirac, Wu-Yang, Banderet), designing a methodology that could be used to examine other monopole and monopole-like models (e.g. the t'Hooft-Polyakov monopole) to determine their physicality.

California State University, Fresno (August 2024-Present)

Project: An entanglement-entropy view of Unruh Temperature and a quantum optics experimental setup to calculate it.

Supervisor: Gerardo Muñoz

Extended the accelerated observer coordinates to all of spacetime to verify the interpretation of the Unruh Effect as a thermal state that arises from entanglement entropy. Determined a possible quantum optics experimental setup involving mechanical oscillators to measure its Unruh temperature, with the predicted Unruh temperature calculated to being within the measurable range for typical quantum optics experiments. This project is now focused on analyzing the connection between the entanglement entropy of these mechanical oscillators and Bekenstein-Hawking entropy (which is a distinguishable project from Srednicki's seminal work with coupled harmonic oscillators).

PUBLICATIONS

- [1] Isaiah Ruz and Douglas Singleton. “Electromagnetic field momentum in theoretical magnetic monopole-like models”. In: *American Journal of Physics* 93.6 (June 2025), pp. 467–476. ISSN: 1943-2909. DOI: [10.1119/5.0244726](https://doi.org/10.1119/5.0244726). URL: <http://dx.doi.org/10.1119/5.0244726>.

(In Progress) Isaiah Ruz and Gerardo Muñoz. “Measuring Unruh Temperatures”

CONFERENCE TALKS

Presented my talk “Hidden Momentum and Field momentum in the Wu-Yang Monopole Model” at APS Far West Section 2025

SCHOLARSHIPS AND AWARDS

Jon R. Dews Physics Research Scholarship	May 2025
Harry C. Burbridge Scholarship in Physics	May 2025
Fresno State Graduate Equity Fellowship	Dec 2025

ADDITIONAL SKILLS AND BACKGROUND

In addition to the physics and mathematics courses listed on my transcript, I have self-studied several textbooks to bolster my physics and mathematical knowledge:

- Peskin and Schroeder and “Quantum Field Theory and the Standard Model” by Matthew D. Schwartz
- “Spacetime and Geometry: An Introduction to General Relativity” by Sean Carroll, “General Relativity” by Wald, and “The Large-Scale Structure of Spacetime” by Hawking and Ellis
- Introductory string theory texts such as Polchinski and “String Theory and M-Theory: A Modern Introduction” by Becker, Becker, Schwarz
- Differential Geometry through “Introduction to Topological Manifolds”, “Introduction to Smooth Manifolds”, and “Introduction to Riemannian Manifolds” by John M. Lee
- Algebraic Topology from Hatcher
- “Differential Forms in Algebraic Topology” by Bott and Tu